
Unit – 3 : Session – 12 : SLO – 2 | Necessary Conditions for Deadlock — Fill in the Blanks

Deadlock

1. Mutual exclusion means only one process at a time can use a resource.
2. A process holding at least one resource and waiting to acquire additional resources held by others is said to be in hold and wait condition.
3. In the no preemption condition, resources cannot be forcibly taken from processes; they must release them voluntarily.
4. The condition where each process is waiting for a resource held by the next process in a circular chain is called circular wait.
5. Deadlock occurs if all four conditions (mutual exclusion, hold and wait, no preemption and circular wait) occur simultaneously.
6. A request edge from process P_i to resource R_j indicates P_i has requested a unit of R_j .
7. An assignment edge from resource R_j to process P_i indicates R_j is assigned to P_i .
8. In a Resource Allocation Graph, the set of processes is represented as circles and the set of resource types as rectangles.
9. If a resource allocation graph contains no cycles, then there is no deadlock.
10. If a cycle exists in a resource allocation graph where each resource type has only one instance, it implies deadlock.